Reductions and NP-completeness

Exercise sheet 9

Exercise 9.1. Recall that $L^* = \{w_1 \cdots w_r \mid r \geq 0,\ w_i \in L\}$. Suppose $L \in \text{NP}$. Is it true that $L^* \in \text{NP}$? Prove your answer.

Exercise 9.2. Prove that INDEPENDENT-SET is NP-complete, where

Input: A finite undirected graph G and a non-negative integer k encoded in binary.

Question: Does G contain a set of at least k vertices with no edge between any two of them?

Exercise 9.3. Prove that VERTEX-COVER is NP-complete, where

Input: A finite undirected graph G and a non-negative integer k encoded in binary.

Question: Does G have a set of at most k vertices that contains at least one endpoint of every edge?

Exercise 9.4. Prove that HITTING-SET is NP-complete, where

Input: A finite set U and a family $S_1, \ldots, S_m \subseteq U$, with the elements and sets explicitly listed, and a non-negative integer k encoded in binary.

Question: Is there a set $H \subseteq U$ with $|H| \leq k$ such that $H \cap S_i \neq \emptyset$ for every $i \in \{1, \ldots, m\}$?

Exercise 9.5. Prove that DOUBLE-SAT is NP-complete, where

Input: A Boolean formula φ, encoded as a word.

Question: Does φ have at least two distinct satisfying assignments?

Exercise 9.6. Prove that 3SAT is NP-complete, where

Input: A CNF formula with exactly three literal occurrences in every clause. A literal may occur more than once in a clause.

Question: Is the formula satisfiable?

Exercise 9.7. * Prove that SUBSET-SUM is NP-complete, where

Input: A list of non-negative integers $a_1, \ldots, a_m$ and a non-negative integer t. All integers are encoded in binary.

Question: Is there an index set $I \subseteq \{1, \ldots, m\}$ such that

$$\sum_{i \in I} a_i = t?$$

Exercise 9.8. * Prove that 3COLOUR is NP-complete, where

Input: A finite undirected graph G.

Question: Can the vertices of G be coloured with at most three colours so that the endpoints of every edge have different colours?

Exercise 9.9. * Prove that HAMILTONIAN-CYCLE is NP-complete, where

Input: A finite undirected graph G.

Question: Does G have a cycle that visits every vertex exactly once before returning to its starting vertex?

Exercise 9.10. * Prove that KNAPSACK is NP-complete, where

Input: A list of pairs $(w_1, v_1), \ldots, (w_n, v_n)$ of non-negative integers, and non-negative integers W and V. All integers are encoded in binary.

Question: Is there an index set $I \subseteq \{1, \ldots, n\}$ such that

$$\sum_{i \in I} w_i \leq W \qquad \text{and} \qquad \sum_{i \in I} v_i \geq V?$$

Exercise 9.11. * A nondeterministic Turing machine may have several possible transitions for a state and observed symbol. It accepts an input if some computation halts in the accepting state, and rejects if all computations reject. It runs in polynomial time if every computation halts within a polynomial number of steps.

Prove that $L \in \mathrm{NP}$ if and only if a polynomial-time nondeterministic Turing machine decides L.

Exercise 9.12. * Let NEXP be the class of languages decided by exponential-time nondeterministic Turing machines. Prove that if P = NP, then EXP = NEXP.